Project Documentation: Preliminary Project Documents – I

Project Title:- **NagarSetu AI – AI-Powered Smart City Governance & Public Accountability Platform**

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1. Problem Statement

Urban local bodies and municipal corporations in Indian cities receive a very large number of civic complaints every day, covering issues such as potholes, damaged roads, broken streetlights, garbage accumulation, drainage blockages, and unauthorized construction. Citizens currently report such issues through phone calls, in-person visits, generic complaint portals, or social media, which do not provide a structured or trackable process.

In most existing systems, complaints are manually read, manually categorized, and manually routed to the concerned department or ward officer. This manual process leads to delays in categorization, incorrect or delayed routing to the responsible authority, duplicate complaints for the same issue from multiple citizens, absence of enforceable resolution deadlines, and a lack of automatic escalation when an issue remains unresolved. Citizens rarely receive status updates and have no reliable way to verify whether a reported issue has actually been fixed, which reduces public trust in the grievance-redressal process.

In addition, municipal administrators lack a consolidated, data-driven view of complaint volumes, department-wise performance, and ward-wise service quality, which makes it difficult to plan preventive maintenance or identify recurring problem areas.

Therefore, there is a need to develop an AI-assisted, web-based Civic Issue Reporting and Resolution platform that automatically classifies and prioritizes citizen complaints, routes them to the correct municipal authority, detects duplicate reports of the same issue, enforces resolution deadlines with automatic escalation, and provides transparent, verifiable tracking for both citizens and administrators.

The proposed system, NagarSetu AI, will allow citizens to report civic issues with photographs and location details, will use AI models to classify, prioritize, and group duplicate complaints, and will route each complaint through a configurable municipal workflow (ward authority → officer → department → worker) with SLA-based deadlines and automatic escalation for unresolved issues.

The system aims to reduce manual effort in complaint handling, improve the speed and accuracy of issue routing, ensure accountability through deadline tracking and proof-of-work verification, and provide municipal administrators with performance analytics to support better civic governance.

Problem in Brief

1. Manual reading and categorization of citizen complaints by municipal staff.
2. Incorrect or delayed routing of complaints to the responsible department or officer.
3. Multiple citizens independently reporting the same civic issue, creating duplicate cases.
4. Absence of clearly defined and enforced resolution deadlines (SLAs) for different complaint types.
5. No automatic escalation mechanism when a complaint remains unresolved beyond its deadline.
6. Lack of verifiable proof that a reported issue has actually been resolved before a complaint is closed.
7. Limited visibility for citizens into the status and history of their complaint.
8. Absence of consolidated, ward-wise and department-wise performance data for administrators.

Proposed Solution

The proposed NagarSetu AI platform will provide a centralized web application through which citizens can report civic issues with an image, description, and location, and through which municipal staff can manage, track, and resolve these issues in a structured, role-based workflow.

The system will use AI/ML models to automatically classify the category and department for each complaint, estimate a severity/priority score, detect duplicate complaints referring to the same civic issue, and assist in verifying whether uploaded before/after images indicate that work has actually been completed. Each complaint will be assigned an SLA-based deadline; complaints not resolved within the deadline will be automatically escalated to the next authority level and flagged in the system.

Municipal workers will update complaint status and upload proof-of-work images, and citizens will be able to verify and rate the resolution before a complaint is marked closed. Administrators will have access to a dashboard showing complaint volumes, SLA compliance, and ward/department-wise performance to support monitoring and decision-making.

The proposed system is expected to reduce manual complaint-handling effort, speed up issue routing, improve accountability through deadline enforcement and verifiable proof of work, and give municipal administrators data-driven visibility into civic service performance.

2. Target Users

NagarSetu AI is intended for municipal corporations and urban local bodies that need to manage citizen civic complaints, coordinate municipal staff, and monitor service performance in a structured, accountable manner. For the MCA Semester III prototype, the system will be configured for a sample municipal area (Ichalkaranji) with representative ward data.

The major target users of the system are as follows:

2.1 Citizen

A resident of the city who reports civic issues (such as potholes, garbage, broken streetlights, or drainage problems) and tracks their resolution.

Major Activities

- Register and maintain a basic profile with contact details and area/address.
- Report a civic issue with a photo/video, description, and location.
- Track the status and timeline of a submitted complaint.
- Receive notifications about assignment, escalation, and completion of the complaint.
- Verify whether the reported issue has actually been resolved and rate the resolution.
- Reopen a complaint if the issue has not genuinely been fixed.

2.2 Ward Authority / Municipal Officer

The Ward Authority or Municipal Officer is responsible for reviewing complaints assigned to their ward or department and forwarding them for action.

Major Activities

- Review AI-classified complaints assigned to their ward.
- Accept, reassign, or reject a complaint with a reason.
- Assign accepted complaints to the concerned department/supervisor.
- Monitor SLA deadlines and pending complaints in their ward.
- View ward-level complaint and performance reports.

2.3 Department Supervisor / Municipal Worker

Department Supervisors assign work orders to municipal workers, and workers carry out and record the actual field resolution of complaints.

Major Activities

- Receive and review complaints assigned to the department.
- Create work orders and assign them to available workers.
- Mark work as started and later as completed.
- Upload before/during/after proof-of-work images for each complaint.
- Update the status of assigned work orders.

2.4 City/Municipal Administrator

The Administrator manages system configuration and requires consolidated information for monitoring and decision-making across the city.

Major Activities

- Manage user accounts, roles, and permissions.
- Configure wards, departments, complaint categories, and SLA rules.
- Monitor city-wide, ward-wise, and department-wise complaint statistics.
- View resolution-rate, SLA-compliance, and citizen-rating reports.
- Identify wards or departments with high pending/escalated complaint counts.

2.5 Target User Summary

Sr. No.	Target User	Major Purpose
1	Citizen	Report civic issues and verify their resolution
2	Ward Authority / Municipal Officer	Review and route complaints within a ward
3	Department Supervisor / Municipal Worker	Execute and record resolution of assigned complaints
4	City/Municipal Administrator	Configure the system and monitor overall performance

The system will implement role-based access control (RBAC) so that each category of user can access only the functions relevant to their responsibilities.

3. Scope

3.1 Scope of the Project

The scope of the proposed NagarSetu AI system covers the design and development of a web-based, AI-assisted platform for reporting, classifying, routing, and tracking the resolution of citizen civic complaints for a municipal area, using Ichalkaranji as a sample/reference deployment with configurable ward and department data.

The system will provide facilities for citizen complaint submission, AI-based classification and duplicate detection, ward/department-based workflow with SLA and escalation, worker proof-of-work tracking, citizen verification, and performance reporting.

3.2 Functional Scope

The proposed system will include the following major functionalities:

1. User & Role Management

- Citizen registration and authentication.
- Role-based accounts for Ward Authority, Department Supervisor, Worker, and Administrator.
- Role-based access control (RBAC).

2. Complaint Submission

- Report a civic issue with an image/video, description, and location (GPS/manual).
- Select from a configurable list of complaint categories (infrastructure, environment, administration, public safety, etc.).

3. AI Classification & Prioritization

- Automatic category/sub-category and department suggestion using a text/image classification model.
- Automatic severity/priority scoring based on complaint attributes.
- Duplicate-complaint detection to group multiple reports of the same issue into one civic incident.

4. Ward Assignment & Workflow

- Automatic mapping of a complaint's location to a configured ward/area.
- Routing of complaints through Ward Authority → Department → Supervisor → Worker.
- Work-order creation and assignment to municipal workers.

5. SLA & Escalation Management

- Configurable SLA (deadline) per complaint category/severity.
- Automatic escalation to the next authority level when an SLA is violated.
- Notifications for assignment, approaching deadline, and escalation.

6. Proof-of-Work & Verification

- Upload of before/during/after images by the assigned worker.
- AI-assisted comparison of before/after images as decision support (not final proof).
- Citizen verification (resolved / partially resolved / not resolved) with rating and feedback.
- Reopening of a complaint if the citizen marks it as not resolved.

7.Dashboard & Reporting

- Complaint timeline/audit trail for each complaint.
- Administrator dashboard with city-wide, ward-wise, and department-wise statistics (total, resolved, pending, escalated, SLA compliance, average resolution time, citizen rating).
- Interactive map view of complaints with status/severity-based filters.

8.Database Management

The system will maintain complaint, user, ward, department, SLA, work-order, and feedback information in a centralized MongoDB database to support consistency, retrieval, and reporting.

3.3 User Scope

The system will support different categories of users with different access permissions, such as:

- Citizen
- Ward Authority / Municipal Officer
- Department Supervisor / Municipal Worker
- City/Municipal Administrator

3.4 Technical Scope

The project will be developed using the MERN stack:

- Frontend: React.js, React Router, Tailwind CSS, Axios, Recharts, React Leaflet, React Hook Form.
- Backend: Node.js with Express.js (REST APIs, authentication, SLA/escalation engine, notifications).
- Database: MongoDB (collections for users, complaints, wards, departments, SLA rules, work orders, proof-of-work, feedback, and audit logs).
- AI/ML components: text and image classification for category/severity prediction and duplicate detection, implemented as a separate service or via a suitable ML/AI API integrated with the backend.
- Development environment: appropriate IDE and version control (Git); deployment on a local server or a free/approved cloud hosting environment for demonstration.

The specific AI/ML approach and hosting platform will be finalized and justified in the project report based on feasibility.

3.5 Out of Scope

The following functionalities are outside the scope of the proposed project:

1. Direct integration with actual police or government databases (a mock/simulated police-coordination module will be used for the prototype).
2. Real-time SMS/WhatsApp integration unless a free/approved gateway is available; otherwise these will be simulated as in-app/email notifications.
3. Legally binding determination of corruption or misconduct; the system will only flag suspicious administrative patterns for human review.
4. Online payment processing and contractor/tender management.
5. Full-scale predictive analytics (complaint forecasting, infrastructure-risk prediction) beyond a basic, demonstrable module.

6 Integration with third-party enterprise or ERP systems.

7 Hardware-based sensors, IoT devices, or CCTV integration.

3.6 Scope Limitation

The project will focus primarily on citizen complaint reporting, AI-assisted classification and duplicate detection, ward/department-based workflow with SLA and escalation, proof-of-work verification, and administrative performance reporting for a sample municipal area. Any additional functionality not mentioned in the approved scope (such as advanced predictive analytics, real police-system integration, or multi-city deployment) will be considered only after discussion with the project guide and based on the available time, resources, and project requirements.

The scope may be refined during the development phase based on feasibility and project-guide recommendations.